Machine Learning Approaches to Predicting Digital Dependence

Sadman Hossain
Department of Computer Science and Engineering
ID: 22299041
BRAC University
Dhaka, Bangladesh
sadman.hossain1@g.bracu.ac.bd

Mahir Asaf Khan
Department of Computer Science and Engineering
ID: 22299187
BRAC University
Dhaka, Bangladesh
mahir.asaf.khan1@g.bracu.ac.bd

***Abstract*—Abstract—This paper presents an empirical machine learning study predicting an individual's Digital Dependence Score from a multivariate dataset of 3,500 records encompassing digital habits, demographics, and validated mental health indicators. The study contributes a complete pipeline: IQR-based outlier treatment, SMOTE-based class balancing, five domain-engineered features, and evaluation of seven regression architectures — Linear Regression, Bayesian Ridge, MLP, Keras DNN, Random Forest, AdaBoost, and KNN. Linear Regression and Bayesian Ridge achieved the highest R^2 of 0.9860 and 0.9859, confirming a predominantly linear feature-to-target mapping. SHAP, LIME, and Partial Dependence Plots identified *device_hours_per_day*, *phone_unlocks*, and *notifications_per_day* as primary predictors, reinforcing the call for transparent, multimodal approaches in digital wellness research.**

***Index Terms*—digital addiction, machine learning, mental health, explainable AI, SHAP, LIME, partial dependence plots, random forest, feature engineering, digital phenotyping**

I. Introduction

The ubiquity of smartphones and social media has fundamentally altered human behaviour, with compulsive device use emerging as a measurable and clinically significant concern. Digital addiction — characterised by persistent, uncontrollable engagement with digital platforms despite adverse consequences to well-being — is increasingly studied through data-driven approaches [2], [3].

This study makes two primary contributions. First, it operationalises a comprehensive ML regression pipeline on 3,500 individuals, evaluating seven architectures across a feature space integrating demographic, behavioural, and clinical mental health data. Second, it applies three complementary XAI techniques — SHAP, LIME, and Partial Dependence Plots — to provide model-level and instance-level interpretability essential for clinical decision support. Section II reviews the literature, Section III details the methodology, Sections IV–V describe the models and XAI analyses, and Section VI presents results.

II. Literature Review

A. Digital Addiction: Theoretical Foundations and Behavioural Predictors

Digital addiction is widely recognised as a significant public health concern characterised by compulsive engagement with digital platforms despite negative consequences. Researchers commonly ground this phenomenon in behavioural addiction theory, which holds that algorithm-driven notifications and social rewards trigger the brain's reinforcement system [2], while Self-Determination Theory further argues that dependencies emerge when platforms are used to satisfy unmet psychological needs for competence and relatedness [3]. Complementary mechanisms such as Fear of Missing Out (FOMO) and social comparison amplify these effects, contributing to heightened anxiety, depression, and diminished well-being [4].

Among the behavioural predictors most consistently identified in the literature, screen time exceeding four to six hours daily, social media checks occurring more than twenty to thirty times per day, and late-night device use — which suppresses melatonin and disrupts circadian rhythms — are strongly associated with high-risk profiles and psychological distress [2], [3]. Critically, screen time alone has been shown to be a poor predictor of mental well-being, with prediction accuracies of only approximately 33%, underscoring that psychological, social, and environmental variables must be incorporated to improve model validity [6]. Protective factors such as regular physical activity and in-person social interaction have been identified as meaningful buffers against addictive digital habits [4]. The present study directly addresses these gaps by constructing a multimodal feature set combining device engagement metrics with validated mental health scores and demographic variables.

B. Machine Learning Approaches to Modelling Digital Behaviour and Mental Health

A growing body of research has applied machine learning to model the relationship between digital behaviours and mental health outcomes. Advanced models including CatBoost, XGBoost, and Neural Networks have demonstrated strong classification performance — achieving precision rates of 85.4% and ROC-AUC scores of 0.93, and accuracy rates of up to 91% in distinguishing addicted from non-addicted individuals [2], [4]. Structural Equation Modelling has further revealed that anxiety and depression mediate the relationship between digital habits and addiction risk [4], while K-means

clustering has enabled the construction of behavioural risk profiles suited to targeted intervention [3].

Passive smartphone sensing represents a particularly promising avenue: a study of 224 college students collecting 44,381 momentary stress observations found that prolonged use of messenger and social network applications, alongside proxies for sleep onset and duration, were the strongest digital markers of stress, though person-specific models yielded only weak correlations (median $r = 0.15$–0.19) and rarely outperformed naive baselines [7]. A 2020 review of 44 passive sensing studies similarly found that no single algorithm consistently outperformed others and identified substantial methodological heterogeneity in feature extraction and evaluation protocols [8]. A scoping review of 65 lifestyle-domain machine learning studies reinforced these concerns, noting that random forest was the most commonly used algorithm and that 74% of studies integrated multi-domain data, yet single-item measurements and unreported data splitting strategies undermined validity [9].

C. Explainability, Personalisation, and Intervention Design

Explainability and intervention design have emerged as increasingly important priorities in this literature. Explainable AI methods, particularly SHAP values, have been applied across multiple studies to identify which features most strongly predict addiction risk or mental health outcomes at both group and individual levels [7], [9], supporting the design of personalised rather than generic interventions. Intervention-focused literature consistently finds that generic advice is insufficient given the highly personalised nature of digital addiction, which varies across personality traits and social contexts [4]. Effective approaches range from AI-driven real-time prompts during emotional distress, to structured digital detox programmes for high-risk users, to digital hygiene curricula that promote balanced engagement rather than total abstinence [2]–[4]. Collectively, the literature points toward the necessity of multimodal, longitudinal data collection, standardised measurement instruments, and transparent reporting practices [8], [9].

III. Methodology

A. Dataset Description

The dataset comprises 3,500 records representing individuals from diverse geographic regions and demographic backgrounds. The feature space contains 24 variables organised into four thematic groups:

- **Demographics:** Age, Gender, Region, Income Level, Education Level, Daily Role.
- **Digital Habits:** Device Hours per Day, Phone Unlocks, Notifications per Day, Social Media Minutes, Study Minutes, Device Type.
- **Mental Health & Lifestyle:** Sleep Hours, Sleep Quality, Physical Activity Days, Anxiety Score, Depression Score, Stress Level, Happiness Score, Focus Score, Productivity Score, High-Risk Flag.
- **Target Variable:** Digital Dependence Score (continuous); mean = 36.68, std = 14.12, range: 5.60–89.20.

A data audit confirmed zero missing values and zero duplicate records across all 3,500 entries. The dataset contained 6 categorical, 7 integer, and 11 floating-point features.

B. Exploratory Data Analysis

Histograms with 30 bins were plotted for all 18 numerical features (Fig. 1). Features such as *phone_unlocks*, *notifications_per_day*, and *social_media_mins* exhibited right-skewed distributions, indicating the presence of heavy users at the upper tail. In contrast, *sleep_quality* and *happiness_score* followed approximately normal distributions. The target variable *digital_dependence_score* was approximately symmetric with a slight right skew at extreme values, validating the suitability of regression-based prediction without target transformation (Fig. 2).

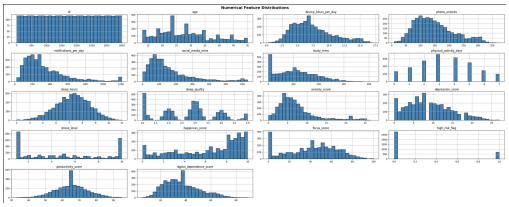


Fig. 1. Numerical feature distributions. Right-skewed distributions are visible for engagement-heavy variables such as *phone_unlocks*, *notifications_per_day*, and *social_media_mins*, indicating heavy-user outlier populations.

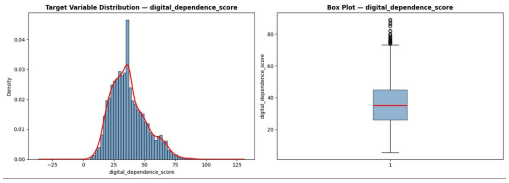


Fig. 2. Distribution of the Digital Dependence Score target variable. The near-symmetric histogram and KDE overlay confirm the suitability of linear regression without target transformation.

The high_risk_flag binary variable exhibited a substantial class imbalance: 2,795 instances (79.86%) belonged to the low-risk class and only 705 (20.14%) to the high-risk class. SMOTE was applied, synthetically generating minority-class samples until both classes reached equal representation. After resampling, both classes contained 2,795 instances each (Fig. 3). High-risk rate analysis across categorical subgroups revealed that certain device types and income brackets exhibited notably elevated proportions of high-risk individuals, motivating targeted digital wellness interventions.

C. Data Preprocessing

Outlier Treatment: Two complementary outlier detection methods were applied. The IQR method identified the most outliers in *anxiety_score* (266), *social_media_mins* (238), and *notifications_per_day* (189). IQR-based clipping was uniformly applied to all numerical columns at $[Q1 - 1.5 \cdot$

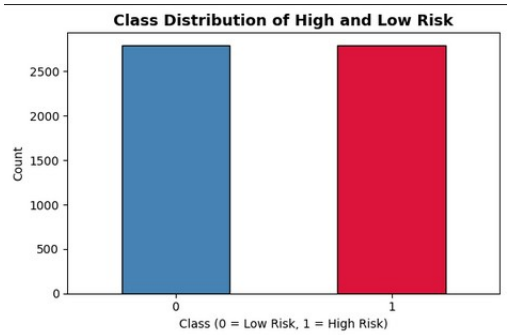


Fig. 3. Class imbalance in *high_risk_flag* before/after SMOTE (left). High-risk rate stratified by categorical subgroups (right), revealing heterogeneous risk across device types and income levels.

IQR, Q3+1.5·IQR], preserving data volume while suppressing extreme observations (Fig. 4).

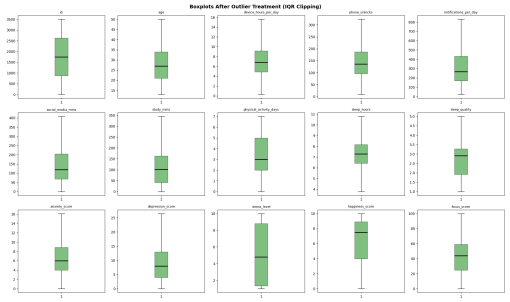


Fig. 4. Boxplots after outlier treatment. Whisker-bound outliers are visibly reduced across engagement and mental health metrics, confirming the effectiveness of IQR clipping.

SMOTE: Applied to *high_risk_flag* exclusively for auxiliary classification analyses; it did not influence the primary regression pipeline. **Categorical Encoding:** Six categorical variables were One-Hot Encoded, expanding the design matrix from 22 to 41 features. **Feature Scaling:** StandardScaler was applied prior to distance-sensitive and gradient-based models (MLP, KNN). **Train-Test Split:** 80% training ($n = 2{,}800$) and 20% test ($n = 700$) with `random_state=42`.

D. Feature Engineering

Five domain-informed composite features were constructed:

- **Mental Health Index:** (happiness + focus + productivity) − (stress + anxiety + depression). Mean = 92.59, std = 29.01, range: 4.31–172.30.
- **Sleep-Stress Ratio:** sleep_quality / (stress_level + 0.01). Mean = 1.13, std = 1.18.
- **Notification Density:** notifications_per_day / (device_hours + 0.01). Mean = 55.33 notifications/hour.
- **Age Group:** Four ordinal bins: ≤24 ($n = 1{,}491$), 25–34 ($n = 1{,}218$), 35–44 ($n = 550$), 45+ ($n = 241$).
- **Heavy User Flag:** Binary indicator for device_hours > 10, identifying 681 individuals (19.46%).

After feature engineering and label encoding of *age_group*, the final design matrix comprised 29 features for neural network inputs.

E. Correlation Analysis

A Pearson correlation matrix was computed on all numerical variables (Fig. 5). The analysis revealed strong negative correlations between *digital_dependence_score* and *happiness_score*, and moderate positive correlations with *anxiety_score* and *device_hours_per_day*. Notable multicollinearity was identified among *anxiety_score*, *depression_score*, and *stress_level*, confirming the suitability of regularised models (Bayesian Ridge) alongside standard linear regression.

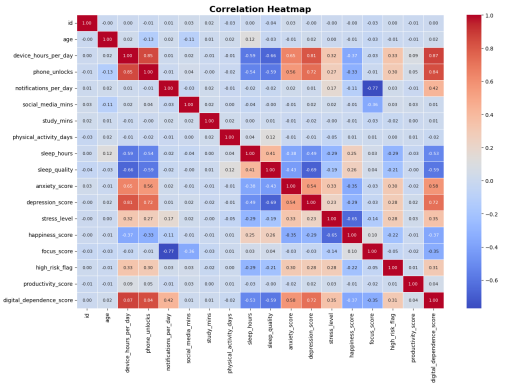


Fig. 5. Correlation heatmap. Digital dependence exhibits positive correlations with device engagement metrics and negative correlations with happiness and sleep quality. Multicollinearity among anxiety, depression, and stress scores is clearly visible.

IV. Machine Learning Models

Seven regression architectures were trained and evaluated, spanning traditional statistical estimators, ensemble methods, and deep neural networks.

Linear Regression: OLS with no regularisation on the 41-feature OHE matrix achieved R^2 = 0.9860 and MSE = 2.80 — the highest performance across all models, confirming a predominantly linear feature-to-target mapping.

Bayesian Ridge: Probabilistic framework treating regression coefficients as Gaussian random variables, yielding adaptive L2 regularisation. Achieved R^2 = 0.9859 and MSE = 2.80, with improved noise robustness appropriate for self-reported psychometric data.

Multilayer Perceptron (MLP): Two implementations were evaluated. The scikit-learn MLP (128→64 neurons, ReLU, Adam, `max_iter=500`, 29-feature matrix) achieved R^2 = 0.9416. The Keras DNN (128→Dropout(0.2)→64→32→1, Adam, MSE loss, 50 epochs, `batch_size=32`) achieved R^2 = 0.9656. Both variants underperformed the linear models, indicating that marginal non-linear complexity is not justified by the dataset's predominantly linear feature structure.

Random Forest: Bagging ensemble of 100 decision trees (`max_depth=5`, `bootstrap=True`, `random_state=42`) achieved test R^2 = 0.9556, MSE =

8.89, and five-fold CV R^2 = 0.9441±0.0033, demonstrating excellent generalisation stability.

AdaBoost: Sequential boosting ensemble (`n_estimators=50`, `learning_rate=0.5`, decision tree stumps) achieved R^2 = 0.9045, MSE = 19.13, and CV R^2 = 0.8996±0.0027.

K-Nearest Neighbors: Non-parametric method ($k = 5$, Euclidean distance, StandardScaler) achieved R^2 = 0.7166 and MSE = 56.70 — the lowest performance, attributable to the curse of dimensionality in the 41-dimensional OHE feature space.

V. Explainable AI (XAI) Analysis

In accordance with the literature's call for transparent and interpretable models in digital health [9], three complementary XAI techniques were applied: SHAP, LIME, and Partial Dependence Plots. Table I summarises the full set of techniques applied.

TABLE I
XAI Techniques — Models, Scope, and Outputs

Technique	Model	Scope	Output
SHAP (Tree)	Random Forest	Global+Local	Beeswarm, Bar, Waterfall, Force, Dependence
SHAP (Kernel)	AdaBoost	Global	Beeswarm summary
LIME	RF / ADA / MLP	Local	Feature weights (15 features)
PDP	Random Forest	Global	Marginal effect curves (Top 6)
PDP + ICE	Random Forest	Global+Local	ICE overlay on PDP

A. SHAP Analysis

SHAP TreeExplainer was applied to Random Forest with a background sample of 100 training instances under interventional feature perturbation. SHAP values were computed for all 700 test instances (SHAP matrix shape: 700×41). The global beeswarm plot (Fig. 6) reveals that *device_hours_per_day*, *notifications_per_day*, and *phone_unlocks* are the three most influential predictors, with high feature values consistently pushing predictions upward. Mental health metrics — particularly *anxiety_score* and *depression_score* — exhibit strong negative SHAP contributions at low values, corroborating the inverse relationship between psychological well-being and digital dependence [3], [4].

The mean absolute SHAP bar chart confirms *device_hours_per_day* as the dominant predictor with a mean |SHAP| approximately three times that of the second-ranked feature [2]. For the first test instance (actual = 32.60, predicted = 30.61), the SHAP waterfall plot (Fig. 7) decomposes the prediction into additive feature contributions from the model's global expected value.

SHAP dependence plots for *device_hours_per_day* and *notifications_per_day* (Fig. 8) reveal monotonically increasing SHAP contributions, confirming dose-response relationships

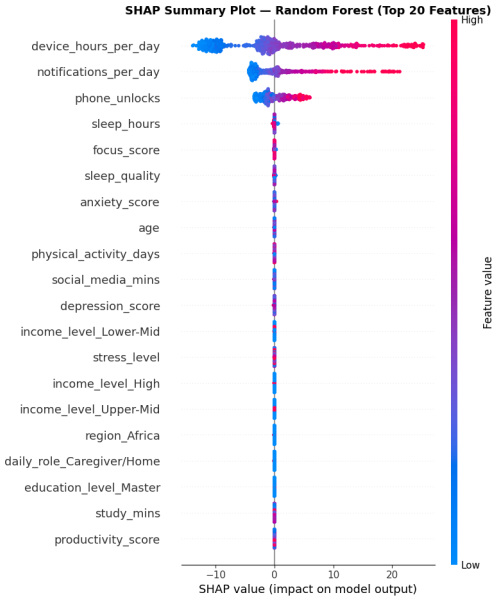


Fig. 6. SHAP beeswarm plot for the Random Forest model. High values of *device_hours_per_day* (red) consistently increase predicted dependence, while strong mental health scores reduce it.

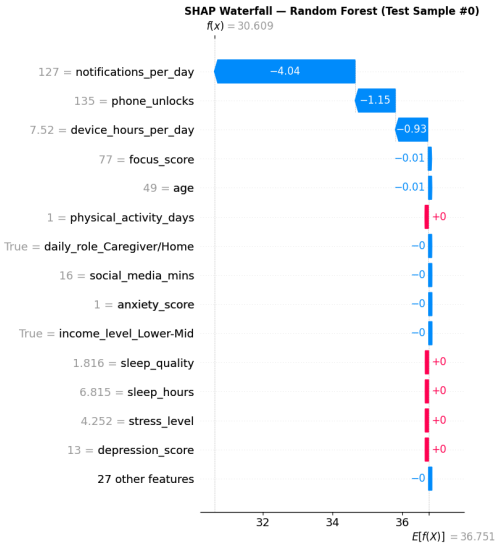


Fig. 7. SHAP waterfall plot for a single prediction (actual=32.60, predicted=30.61). Each bar quantifies one feature's additive contribution to the final prediction from the global base value.

directly applicable to clinical threshold-setting. SHAP KernelExplainer applied to AdaBoost (20-instance background, 50-instance test subset) demonstrated broad cross-model agreement in feature importance orderings despite the models' fundamentally different learning paradigms.

B. LIME Analysis

A `LimeTabularExplainer` was initialised with 2,800 training instances, 41 features, continuous feature discretisation, and regression mode. For Test Sample #0 (actual = 32.60), LIME yielded predictions of 30.61 (Random Forest,

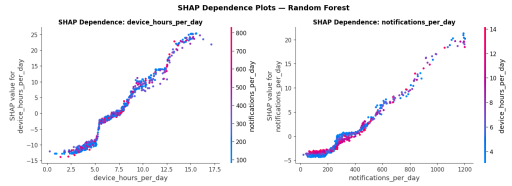


Fig. 8. SHAP dependence plots for the top two predictors. Near-linear trends confirm dose-response relationships between device engagement and digital dependence.

local R^2 = 0.076), 33.20 (AdaBoost, local R^2 = 0.080), and 29.91 (MLP sklearn, local R^2 = 0.190). The consistency of LIME-identified features across all three models supports the predictive primacy of behavioural frequency variables established in the SHAP global analysis [7]. The higher local R^2 for the MLP suggests a more locally linear decision boundary compared to the ensemble models at this instance.

C. Partial Dependence Plots

PDPs were computed for the top six SHAP-identified features: *device_hours_per_day*, *notifications_per_day*, *phone_unlocks*, *sleep_hours*, *focus_score*, and *sleep_quality*. The six-panel PDP (Fig. 9) reveals monotonically increasing marginal dependence for device engagement features, while sleep features exhibit clear negative marginal effects, confirming that restorative behaviours act as protective factors against digital dependence [3]. A PDP + ICE overlay for *device_hours_per_day* reveals heterogeneous individual response curves around the average PDP trend, motivating person-centred digital health interventions rather than population-level threshold policies [4].

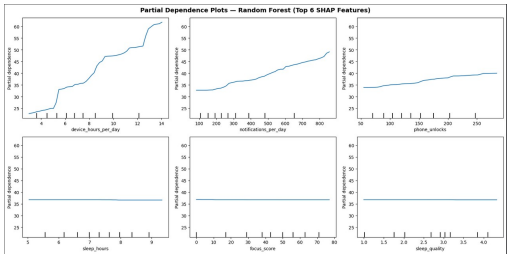


Fig. 9. Six-panel PDP for the Random Forest model. Device engagement metrics show monotonically increasing effects; sleep quality and sleep hours exhibit clear protective (negative) partial dependence.

VI. Results and Discussion

A. Comparative Model Performance

Table II presents the comprehensive performance summary across all seven models.

Linear Regression achieved R^2 = 0.9860 and MSE = 2.80; Bayesian Ridge followed at R^2 = 0.9859 and MSE = 2.80. The negligible difference confirms a predominantly linear feature-to-target relationship. The Bayesian model's adaptive regularisation makes it preferable for deployment on datasets with self-reported measurement error [2]. The Keras DNN (R^2 = 0.9656) outperformed the scikit-learn MLP (R^2 = 0.9416), with the gap attributable to the DNN's deeper architecture, dropout regularisation, and use of the 29-feature engineered matrix. Both neural network variants underperformed linear models, consistent with the observation that black-box models offer no practical advantage over interpretable alternatives when feature-target relationships are approximately linear [9]. Random Forest (CV R^2 = 0.9441±0.0033) demonstrated strong and stable generalisation. AdaBoost (CV = 0.8996±0.0027) performed below Random Forest, consistent with the established finding that boosting approaches are less competitive than bagging in datasets with behavioural outlier variance [4]. KNN (R^2 = 0.7166, MSE = 56.70) demonstrated the poorest performance by a substantial margin due to the curse of dimensionality in the 41-dimensional OHE feature space [9].

TABLE II
Comprehensive Performance Summary — Digital Dependence Score Prediction

Model	R^2	MSE	CV R^2	Rank
Linear Regression	0.9860	2.80	—	1
Bayesian Ridge	0.9859	2.80	—	2
Keras DNN	0.9656	—	—	3
Random Forest	0.9556	8.89	0.9441 ± 0.003	4
MLP (Scikit-learn)	0.9416	—	—	5
AdaBoost	0.9045	19.13	0.8996 ± 0.003	6
K-Nearest Neighbors	0.7166	56.70	—	7

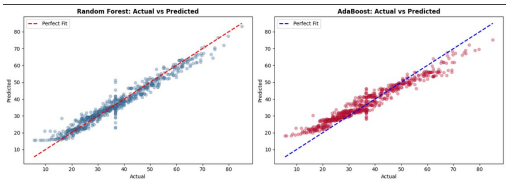


Fig. 10. Actual vs. predicted scatter plots. Both ensemble models closely track the perfect-fit diagonal; Random Forest demonstrates tighter clustering, confirming superior precision.

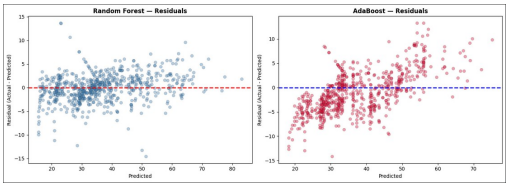


Fig. 11. Residual scatter plots. Random Forest residuals are more symmetrically distributed around zero, indicating better homoscedasticity and lower systematic prediction bias.

B. Feature Importance and XAI Convergence

Both Random Forest Gini-based importance (Fig. 12) and SHAP-based importance converge on the same top three predictors: *device_hours_per_day*, *notifications_per_day*, and *phone_unlocks*. This dual-method confirmation provides robust evidence for the centrality of active device engagement

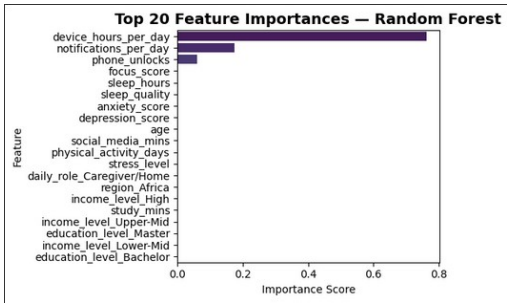


Fig. 12. Random Forest feature importance by mean impurity decrease. Device engagement features dominate, consistent with SHAP-based rankings in Section V.

metrics as the primary drivers of digital dependence prediction. LIME local analyses confirmed instance-level consistency of these drivers, though local R^2 scores (0.076–0.190) indicate that local linear approximations capture a modest portion of instance-level variance — consistent with passive sensing studies that found weak person-specific model correlations [7].

C. Discussion

The experimental results provide a convergent and empirically grounded account of digital dependence predictability from behavioural and psychological biomarkers. The finding that Linear Regression achieves $R^2 = 0.9860$ — outperforming both deep neural networks and ensemble methods — confirms that the dominant variance in digital dependence is captured by direct, additive feature-to-target relationships. This is significant for clinical translation: linear models enable practitioners to directly quantify the expected change in dependence score corresponding to a specific reduction in daily device hours or notification volume.

The XAI analysis provides a multi-layered interpretability framework that directly addresses the literature's call for explainable, personalised digital wellness tools [9]. SHAP global analyses confirm that device engagement metrics are the primary risk drivers, while sleep quality and mental health indicators serve as protective moderators, consistent with behavioural addiction theory [2], [3]. PDP analyses confirm that the protective effects of sleep and the risk-amplifying effects of engagement are monotonic and additive — providing a direct empirical basis for targeted behavioural intervention, such as reducing social media minutes while improving sleep hygiene as a dual-axis strategy. The individual-level heterogeneity revealed by ICE plots reinforces the conclusion that personalised, AI-driven intervention design is both warranted and technically feasible [4].

VII. Conclusion

Across seven regression architectures evaluated on 3,500 records, Linear Regression and Bayesian Ridge achieved the highest performance (R^2 = 0.9860 and 0.9859), confirming a predominantly linear feature-to-target relationship. Three complementary XAI techniques provided convergent interpretability: SHAP identified *device_hours_per_day*, *notifications_per_day*, and *phone_unlocks* as primary predictors; PDPs confirmed monotonic dose-response relationships for engagement metrics and protective effects for sleep variables; and LIME demonstrated instance-level consistency. These findings address key gaps in multimodal feature integration, transparent reasoning, and empirically grounded intervention design [8], [9]. Future work should pursue longitudinal modelling, feature selection via RFE or LASSO, external validation on demographically distinct populations [8], and real-time passive sensing integration for prospective prediction.

Acknowledgment

The authors thank the Department of Computer Science and Engineering at BRAC University for the academic resources and guidance that supported this research. Gratitude is also extended to the open-source communities behind scikit-learn, TensorFlow/Keras, SHAP, LIME, and the pandas/NumPy ecosystem, which formed the technical backbone of this study.

References

[1] E. Ashok, "Can mental health struggles involving brain-body coherence be predicted by a user's screen time and app usage patterns by training a machine learning algorithm to create a regression model?" Poster presentation, Arizona State University, n.d.

[2] D. R. Hussein, A. S. Alhumaima, H. Alkattan, and M. Abotaleb, "Machine learning analysis of social media usage patterns and mental health indicators," *Journal of Transactions in Systems Engineering*, vol. 3, no. 3, pp. 471–488, 2025.

[3] A. O. Ibitoye, S. Ravindran, O. F. Afe, and A. O. Abiodun, "Predicting digital addiction patterns with machine learning for personalised mental health support," *Journal of Social Computing*, vol. 6, no. 4, pp. 359–377, 2025.

[4] N. Lagun and A. Gautam, "Modeling the impact of screen time on mental wellness through computational analysis," *Preprints.org*, 2025.

[5] S. A. Wahed and M. A. Wahed, "AI-driven digital well-being: Developing machine learning model to predict and mitigate internet addiction," *LatIA*, vol. 3, p. 73, 2025.

[6] A. A. Alhur, M. Alsahmmari, and M. A. Al-Khattab, "Can lifestyle habits predict happiness? An exploratory machine learning study using a visual data mining platform," *Journal of Pioneering Medical Sciences*, vol. 14, no. 6, pp. 45–52, 2025.

[7] G. Aalbers, A. T. Hendrickson, M. M. P. Vanden Abeele, and L. Keijsers, "Smartphone-tracked digital markers of momentary subjective stress in college students: Idiographic machine learning analysis," *JMIR mHealth and uHealth*, vol. 11, p. e37469, 2023.

[8] Y. Nishiyama et al., "Smartphone-based mental state estimation," *Journal of Information Processing*, vol. 28, pp. 16–30, 2020.

[9] T. Estrella, L. Capdevila, C. Alfonso, and J.-M. Losilla, "Machine learning for the analysis of healthy lifestyle data: Scoping review and guidelines," *JMIR Human Factors*, vol. 13, p. e78648, 2026.

[10] L. Luo et al., "Mental health issues and 24-hour movement guidelines-based intervention strategies for university students with high-risk social network addiction: Cross-sectional study using a machine learning approach," *Journal of Medical Internet Research*, vol. 27, p. e72260, 2025.